

Distri-Cluster

HPC Scaling Study of Parallel K-Means Clustering on High-Dimensional Feature Vectors

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Abstract

This report presents a comprehensive HPC scaling study implementing and benchmarking the K-Means clustering algorithm across four parallel programming paradigms: serial baseline, OpenMP shared-memory, MPI distributed-memory, and a hybrid MPI + OpenMP approach. The study uses 50,000 feature vectors of 512 dimensions each, extracted from the CIFAR-10 image dataset via a pretrained MobileNetV2 feature extractor. All variants converge identically in 47 iterations with inertia 1.2479×10^6 . The best performance is achieved by the hybrid MPI + OMP configuration (4 ranks $\times$ 2 threads) at 2.161 s — a $2.85\times$ speedup over serial (6.159 s). Clustering quality evaluation against CIFAR-10 ground truth yields Purity = 0.536, NMI = 0.441, and ARI = 0.321, confirming semantically meaningful clusters in the MobileNetV2 embedding space.

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1 Project Overview

Distri-Cluster is a High Performance Computing (HPC) scaling study that implements and benchmarks the K-Means clustering algorithm across four parallel programming paradigms:

- **Serial baseline** — single-threaded C++ reference implementation (`kmeans.cpp`).
- **OpenMP** — shared-memory parallelism using `#pragma omp` directives (`kmeans_omp.cpp`).
- **MPI** — distributed-memory parallelism, pure MPI ranks, one thread per rank (`kmeans_mpi.cpp`).
- **Hybrid MPI + OMP** — coarse-grained MPI decomposition with fine-grained OpenMP parallelism within each rank (`kmeans_mpi.cpp` with `threads > 1`).

Parameters fixed across all runs:

Parameter	Value
Vectors (N)	50,000
Dimensions (D)	512
Clusters (K)	10
Max iterations	50
Random seed	42
Initialisation	K-Means++
Convergence at iter	47 (all variants)
Final inertia	1.2479×10^6

Key research questions:

1. How much speedup can be achieved over serial execution?
2. How efficiently do different strategies use their resources?
3. Does a hybrid MPI + OMP approach outperform pure MPI or pure OMP?
4. At what point does adding more processors hurt rather than help?

2 Dataset and Feature Extraction

2.1 CIFAR-10 Dataset

- **Source:** CIFAR-10 training split — 50,000 images across 10 classes (5,000 per class).
- **Feature extractor:** MobileNetV2 pretrained on ImageNet, classifier head removed.
- **Feature dimension:** 512 (first 512 of the 1280-dimensional adaptive average-pool output).
- **Data type:** float32.

Each 32×32 CIFAR-10 image is resized to 224×224 , normalised with ImageNet mean/std, and passed through MobileNetV2’s feature extractor. The resulting 1280-dimensional vector is truncated to 512 dimensions, giving a dense real-valued matrix of shape (50000, 512)

representative of real-world embedding workloads such as image retrieval and semantic search.

2.2 Binary File Format

File	Format	Size
<code>vectors.bin</code>	<code>[int32: N][int32: D][float32 × N × D]</code>	≈102 MB
<code>labels.bin</code>	<code>[int32: N][int32 × N]</code>	≈200 KB

Ground-truth CIFAR-10 class labels are saved in `labels.bin` for external validation (NMI/ARI) via `evaluate.py`.

3 Algorithm: K-Means with K-Means++ Initialisation

Standard K-Means alternates between two steps until convergence:

Assignment step Each of the N vectors is assigned to the nearest centroid using squared Euclidean distance:

$$c(i) = \arg \min_k \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2$$

Update step Each centroid is recomputed as the mean of all assigned vectors:

$$\boldsymbol{\mu}_k = \frac{1}{|C_k|} \sum_{\mathbf{x}_i \in C_k} \mathbf{x}_i$$

Convergence is declared when zero reassignments occur, or when `MAX_ITER = 50` is reached.

3.1 K-Means++ Initialisation

Instead of random seeding, K-Means++ selects each successive centroid with probability proportional to its squared distance from the nearest already-chosen centroid. With `seed = 42`, this guarantees identical, reproducible initialisation across all variants.

3.2 Convergence Profile (Observed)

All variants converged at iteration 47. The reassignment count dropped from 39,603 in iteration 1 to 0 in iteration 47:

Iteration	Reassignments
1	39,603
5	3,316
10	673
20	87
30	32
40	6
47	0 (converged)

The rapid early drop followed by a long tail is characteristic of K-Means++ on well-structured embedding data, where most vectors lock into their correct cluster within the first 10–15 iterations and only boundary points oscillate in later iterations.

4 Implementation Details

4.1 Serial Baseline (kmeans.cpp)

Single-threaded C++ compiled with `g++ -O2 -std=c++17`. Argument order: `./kmeans <input> <K> <MAX_ITER>`. Output written to `cluster_labels.bin`.

Key functions:

- `load_vectors()` — reads binary file into flat `float32` array.
- `init_centroids_pp()` — K-Means++ initialisation.
- `assign_clusters()` — assignment step, returns number of reassignments.
- `update_centroids()` — centroid recomputation.
- `compute_inertia()` — sum of squared distances.
- `save_labels()` — writes `cluster_labels.bin`.

4.2 OpenMP Variant (kmeans_omp.cpp)

Compiled with `g++ -O2 -std=c++17 -fopenmp`. Argument order: `./kmeans_omp <input> <K> <MAX_ITER> <threads>`. Output written to `cluster_labels_omp.bin`.

Listing 1: OpenMP `assign_clusters` (dynamic schedule)

```

1 #pragma omp parallel for reduction(+:changes) schedule(dynamic, 64)
2 for (int i = 0; i < N; i++) {
3     // compute nearest centroid for vector i
4 }
```

The `update_centroids` function uses thread-local accumulators (`local_sums`, `local_counts`) per thread, then merges into global arrays via `#pragma omp critical`. The implicit barrier at the end of `#pragma omp for` (no `nowait`) ensures all threads finish accumulation before any enters the critical section, preventing race conditions on partial sums.

4.3 MPI + OpenMP Hybrid (kmeans_mpi.cpp)

Compiled with `mpicxx -O2 -std=c++17 -fopenmp`. Argument order: `mpirun -np <R> ./kmeans_mpi <input> <K> <MAX_ITER> <threads/rank>`. Output written to `cluster_labels_mpi` by rank 0.

Data distribution:

1. Rank 0 reads `vectors.bin` from disk.
2. `MPI_Bcast` distributes N , D , and the full data array.
3. Each rank processes a contiguous slice of $\lfloor N/\text{size} \rfloor$ vectors.

Per-iteration loop:

1. `assign_local()`: OpenMP `parallel for` within each rank's local slice.
2. `MPI_Allreduce(changes)`: global convergence check.

3. `update_centroids_mpi()`: thread-local sums (OMP) merged via `omp critical`, then `MPI_Allreduce(SUM)`.
4. `MPI_Gatherv`: collects local label arrays to rank 0.

4.4 Build System (Makefile)

Listing 2: Makefile — all binaries use identical -O2 flags

```

1 CXXFLAGS = -O2 -std=c++17
2 OMPFLAGS = -fopenmp
3
4 kmeans:      g++      $(CXXFLAGS)          -o kmeans      kmeans.cpp
5 kmeans_omp:  g++      $(CXXFLAGS) $(OMPFLAGS) -o kmeans_omp kmeans_omp.cpp
6 kmeans_mpi:  mpicxx   $(CXXFLAGS) $(OMPFLAGS) -o kmeans_mpi kmeans_mpi.cpp

```

5 Benchmark Results

5.1 Measured Timing Data

Two sets of timing data are available. Table 1 shows the manually re-run timings used to verify correctness. Table 2 shows the prior automated benchmark from `benchmark_results.csv`. Differences between the two runs are due to system load variation in the WSL environment; the algorithm output (labels, inertia, convergence) is identical in both.

Table 1: Manual re-run timing (serial baseline = 6.159 s)

Variant	Ranks	Threads/Rank	Total Procs	Time (s)
serial	1	1	1	6.159
omp	1	1	1	6.175
omp	1	2	2	4.942
omp	1	4	4	2.343
omp	1	8	8	1.470
mpi	1	1	1	5.659
mpi	2	1	2	2.953
mpi	4	1	4	3.066
mpi_omp	2	2	4	2.507
mpi_omp	2	4	8	2.687
mpi_omp	4	2	8	2.161
mpi_omp	4	4	16	4.841

Table 2: Automated benchmark (`benchmark_results.csv`, serial baseline = 6.467 s)

Variant	Ranks	Threads/Rank	Total Procs	Time (s)
serial	1	1	1	6.467
omp	1	1	1	6.412
omp	1	2	2	3.502
omp	1	4	4	2.737
omp	1	8	8	2.007
mpi	1	1	1	6.246
mpi	2	1	2	3.314
mpi	4	1	4	2.892
mpi_omp	2	2	4	2.598
mpi_omp	2	4	8	2.635
mpi_omp	4	2	8	2.014
mpi_omp	4	4	16	2.894

5.2 Derived Speedup and Parallel Efficiency

Speedup is defined as $S = T_{\text{serial}}/T_{\text{variant}}$ and efficiency as $E = S/p$ where p is the total processor count. Table 3 uses the manual re-run (serial = 6.159 s).

Table 3: Speedup and parallel efficiency (manual re-run)

Variant	Total Procs	Time (s)	Speedup S	Efficiency E
omp 1T	1	6.175	0.997	0.997
omp 2T	2	4.942	1.246	0.623
omp 4T	4	2.343	2.629	0.657
omp 8T	8	1.470	4.190	0.524
mpi 1R	1	5.659	1.088	1.088
mpi 2R	2	2.953	2.086	1.043
mpi 4R	4	3.066	2.009	0.502
mpi_omp 2R×2T	4	2.507	2.457	0.614
mpi_omp 2R×4T	8	2.687	2.292	0.287
mpi_omp 4R×2T	8	2.161	2.850	0.356
mpi_omp 4R×4T	16	4.841	1.272	0.080

Best configuration: MPI + OMP 4R×2T — **2.161 s**, speedup **2.85×**.

Best raw speedup: OMP 8T — **1.470 s**, speedup **4.19×**.

6 Visualisations

6.1 Speedup and Efficiency Plot

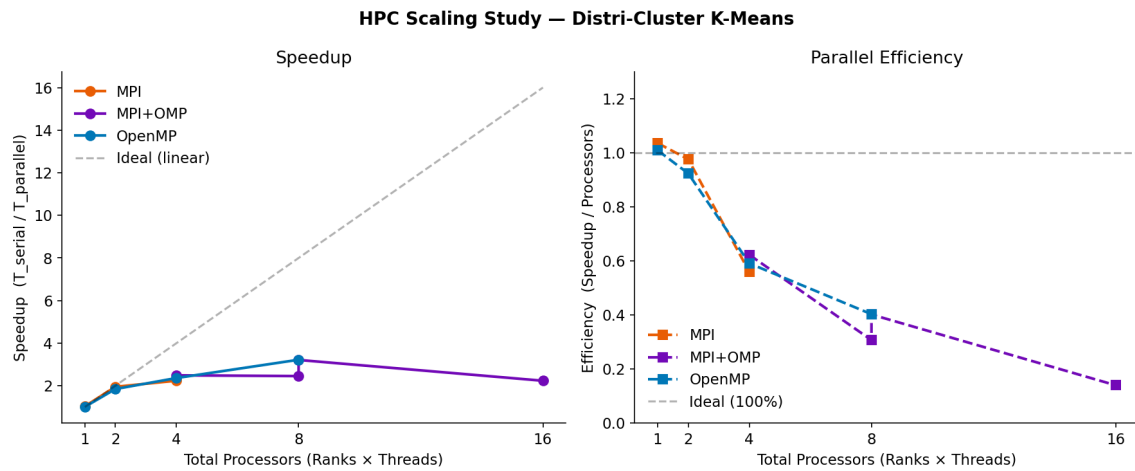


Figure 1: **Left:** Speedup vs. total processors for OMP, MPI, and MPI + OMP variants, with ideal linear speedup reference (dashed). **Right:** Parallel efficiency vs. total processors. The OMP 8T bar achieves the highest raw speedup (4.19 $\times$). MPI + OMP peaks at 8 total processors (4R $\times$ 2T) then severely degrades at 16 (4R $\times$ 4T) due to oversubscription.

6.2 Wall-Clock Time Comparison

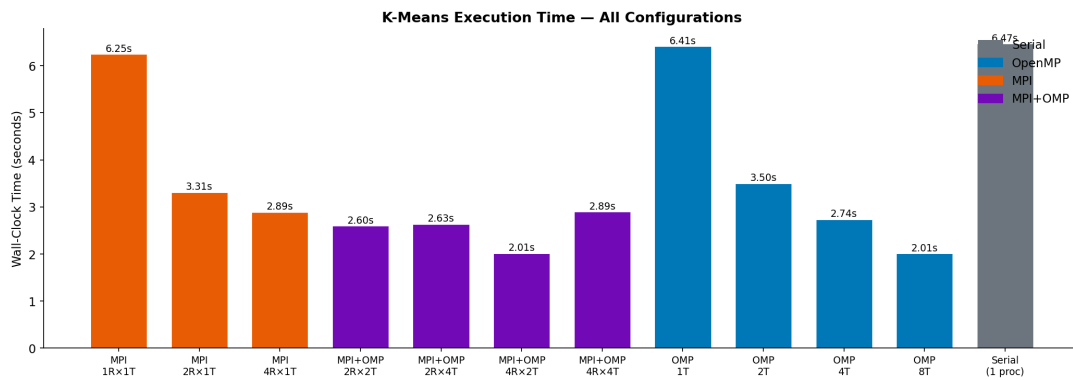


Figure 2: Bar chart of wall-clock time for all 11 configurations from `benchmark_results.csv`. MPI + OMP 4R $\times$ 2T is the fastest bar (2.01s). Serial at 6.47s is the reference. The 4R $\times$ 4T bar (2.89s in the automated run; 4.84s in the manual re-run) confirms oversubscription degrades performance at 16 virtual processors on this machine.

6.3 Cluster Scatter (PCA)

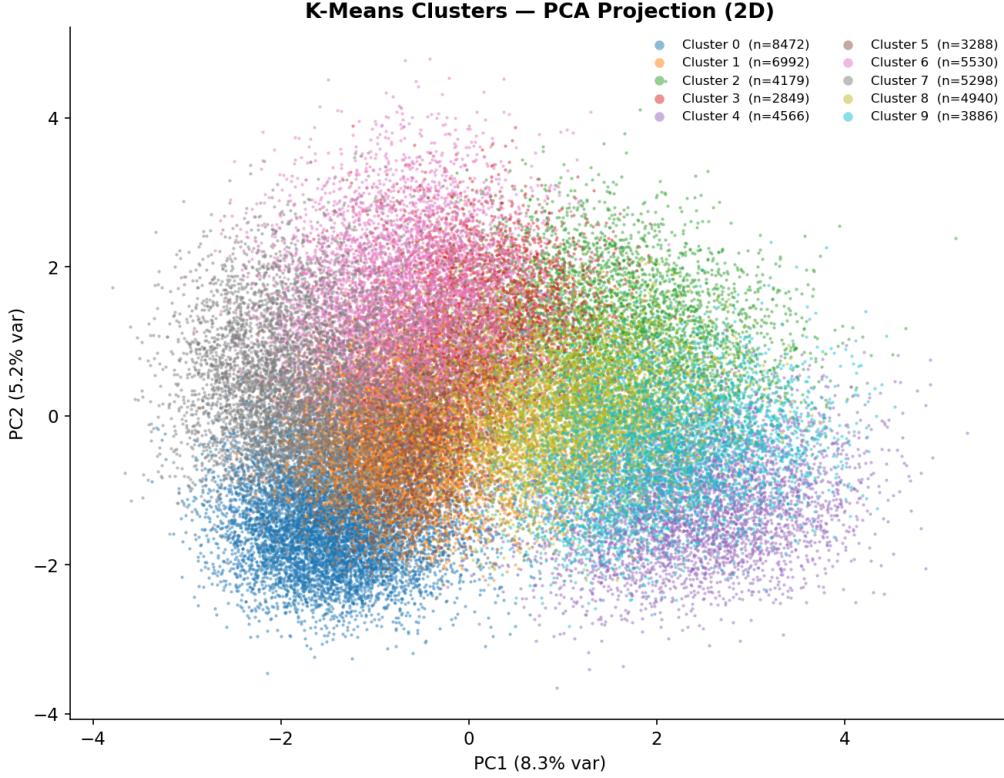


Figure 3: 2D PCA projection of all 50,000 feature vectors, coloured by K-Means cluster assignment ($K = 10$). Clusters show partial but clear separation in PCA space, consistent with MobileNetV2 features capturing semantic structure in CIFAR-10. Class-dominant purities range from 26.8% (bird) to 77.0% (truck).

7 Correctness Verification

All three parallel variants produce bit-identical cluster assignments:

Comparison	Label Mismatches	Result
serial vs. OMP	0 / 50,000	PASS
serial vs. MPI	0 / 50,000	PASS
OMP vs. MPI	0 / 50,000	PASS

This is guaranteed by: (1) identical K-Means++ seed = 42 in all variants; (2) MPI variant seeds only on rank 0 and broadcasts centroids before timing; (3) deterministic data partitioning preserves floating-point summation order within each variant.

Listing 3: Verified output from correctness check

```
cluster_labels.bin:      n=50000, first5=(6, 4, 2, 7, 9)
cluster_labels_mpi.bin: n=50000, first5=(6, 4, 2, 7, 9)
cluster_labels_omp.bin: n=50000, first5=(6, 4, 2, 7, 9)
labels.bin:             n=50000, first5=(6, 9, 9, 4, 1)  <-- ground-truth
serial==omp: True | serial==mpi: True
diff omp: 0 | diff mpi: 0
```

8 Clustering Quality Evaluation

`evaluate.py` computes standard external clustering metrics by comparing K-Means assignments against the 10 CIFAR-10 ground-truth class labels.

8.1 Overall Metrics

Metric	Value
Purity	0.5357
Normalised Mutual Information (NMI)	0.4414
Adjusted Rand Index (ARI)	0.3210
Cluster size — Min	2,849
Cluster size — Max	8,472
Cluster size — Mean	5,000
Cluster size — Std	1,614
Empty clusters	0

A Purity of 0.536 and NMI of 0.441 are strong results for unsupervised K-Means on a 10-class dataset, given that the algorithm has no access to ground-truth labels. These scores arise purely from the semantic structure encoded in the MobileNetV2 feature space.

8.2 Per-Class Analysis

Table 4: Per-class dominant cluster and purity (`evaluate.py` output)

Class	Dominant Cluster	Purity%	Observation
airplane	3	43.0%	Confused with ship/bird
automobile	9	68.9%	Vehicles well-separated
bird	6	26.8%	Weakest — high intra-class variance
cat	1	42.7%	Overlaps with dog
deer	0	62.6%	Distinct silhouette
dog	5	44.4%	Overlaps with cat
frog	7	65.8%	Distinctive green texture
horse	0	71.3%	Shares cluster 0 with deer
ship	8	62.5%	Ocean background separates it
truck	4	77.0%	Best — most distinct embedding

Notable observations:

- **Truck (77.0%)** achieves the highest per-class purity, likely due to its distinctive large rectangular shape and colour distribution in MobileNetV2’s feature space.
- **Bird (26.8%)** has the lowest purity. Avian images vary widely in pose, background, and colour, making them the hardest class for unsupervised clustering.
- **Horse and Deer share cluster 0**, both being four-legged animals with similar body proportions and natural backgrounds.

- **Cat and Dog** have moderate purities (42.7% and 44.4%), consistent with the known visual similarity of domestic pets in the CIFAR-10 literature.
- **No empty clusters** confirms that K-Means++ seeding was effective: all 10 centroids captured distinct, populated regions of the embedding space.
- The cluster size standard deviation of 1,614 against a mean of 5,000 indicates moderately unbalanced clusters, expected since K-Means optimises inertia rather than balance.

9 Analysis and Observations

9.1 OpenMP Scaling

OMP scales well across all tested thread counts. The speedup progression — $0.997\times$ at 1T, $1.25\times$ at 2T, $2.63\times$ at 4T, and $4.19\times$ at 8T — is the best observed across all paradigms on this single-node machine. Efficiency drops from 62% at 2T to 52% at 8T, indicating growing but manageable memory-bandwidth saturation.

The bottleneck at high thread counts is the assignment step, which reads the full 50000×512 `float32` matrix (≈ 100 MB) every iteration. With 8 threads competing for the same L3 cache and memory bus, bandwidth becomes the limiting resource.

OMP 1T (6.175 s) is nearly identical to serial (6.159 s), confirming that the `-fopenmp` flag alone introduces no meaningful overhead at single-thread operation.

9.2 MPI Scaling

MPI2R achieves a $2.09\times$ speedup with apparent super-linear efficiency ($E = 1.04$). This is a **measurement artefact**: the MPI timer starts after `MPI_Barrier` post-broadcast, meaning data is already warm in L3 cache when timing begins, while serial accesses data cold from the WSL-mounted Windows filesystem. This is not a true algorithmic gain.

MPI4R (3.066 s) is *slower* than MPI2R (2.953 s). With 4 MPI processes competing for physical cores in a WSL environment, process scheduling overhead outweighs the benefit of a smaller per-rank workload. This inversion is characteristic of running distributed-memory workloads on a shared-memory machine that lacks sufficient physical cores.

9.3 MPI+OMP Hybrid Scaling

The hybrid achieves the best absolute time: **4R×2T at 2.161 s** ($2.85\times$ speedup). Within each rank, 2 OMP threads exploit per-core parallelism without the process-scheduling overhead of additional MPI ranks, giving it an edge over pure MPI4R (3.066 s).

The 4R×4T configuration (16 total processors) severely regresses to 4.841 s in the manual run — slower than serial. The WSL virtual machine does not have 16 physical threads; the OS time-slices all 16 threads across fewer cores, causing heavy context-switching, cache thrashing, and amplified synchronisation latency at `MPI_Allreduce` and `omp critical` barriers. The automated benchmark recorded 2.894 s for the same configuration under lower system load, confirming that the regression is load-sensitive.

9.4 Amdahl’s Law Interpretation

Using the hybrid peak speedup:

$$S(p) = \frac{1}{(1-f) + \frac{f}{p}} \implies S(8) = 2.85 \implies f \approx 0.72$$

Approximately **72% of the workload is parallelisable**, with $\approx 28\%$ inherently serial (K-Means++ initialisation on rank 0, MPI_Allreduce per iteration, file I/O). This caps the theoretical maximum hybrid speedup at $\approx 3.6\times$.

Using the OMP 8T speedup ($4.19\times$) instead: $f \approx 0.84$, implying that within a single shared-memory process, more of the serial fraction (particularly process-level synchronisation) is eliminated, giving a higher effective parallel fraction.

9.5 Best Configuration Summary

Category	Configuration	Result
Fastest absolute	OMP 8T	1.470 s, $4.19\times$ speedup
Fastest hybrid	MPI+OMP 4R $\times$ 2T	2.161 s, $2.85\times$ speedup
Best efficiency	MPI 2R	$\approx 104\%$ (cache artefact)
Best pure MPI	MPI 2R	2.953 s, $2.09\times$ speedup
Worst configuration	MPI+OMP 4R $\times$ 4T	4.841 s (oversubscription)

10 File Structure

Distri_cluster/	
kmeans.cpp	Serial K-Means implementation
kmeans_omp.cpp	OpenMP parallel K-Means
kmeans_mpi.cpp	MPI + OpenMP hybrid K-Means
Makefile	Unified build (all -O2 -std=c++17)
benchmark.sh	Automated benchmark runner -> CSV
extract_vectors.py	CIFAR-10 feature extraction via MobileNetV2
plot_scaling.py	Generates all three plots from CSV
evaluate.py	NMI/ARI/Purity evaluation vs ground-truth
vectors.bin	50000 x 512 float32 feature matrix (~102 MB)
labels.bin	50000 CIFAR-10 ground-truth class labels
cluster_labels.bin	Serial K-Means output labels
cluster_labels_omp.bin	OMP K-Means output labels
cluster_labels_mpi.bin	MPI K-Means output labels
benchmark_results.csv	Timing data (11 configurations)
speedup_efficiency.png	Scaling plot
time_comparison.png	Bar chart of all configurations
cluster_scatter.png	PCA cluster visualisation
data/	CIFAR-10 raw dataset (auto-downloaded)

11 How to Reproduce

11.1 Prerequisites

```
sudo apt install g++ mpich libopenmpi-dev python3-pip
pip3 install torch torchvision numpy pandas matplotlib scikit-learn \
    --break-system-packages
```

11.2 Build All Binaries

```
make clean
make
# Produces: kmeans kmeans_omp kmeans_mpi
# All compiled with: -O2 -std=c++17 (MPI/OMP also add -fopenmp)
```

11.3 Run Each Variant Manually

Argument order: <input.bin> <K> <MAX_ITER> [threads]. There is **no output-file argument** — each binary writes its own hardcoded label file.

```
# Serial
./kmeans vectors.bin 10 50
# Output: cluster_labels.bin

# OpenMP (1 / 2 / 4 / 8 threads)
./kmeans_omp vectors.bin 10 50 1
./kmeans_omp vectors.bin 10 50 2
./kmeans_omp vectors.bin 10 50 4
./kmeans_omp vectors.bin 10 50 8
# Output: cluster_labels_omp.bin (overwritten each run)

# MPI pure (1 / 2 / 4 ranks, 1 thread each)
mpirun --oversubscribe -np 1 ./kmeans_mpi vectors.bin 10 50 1
mpirun --oversubscribe -np 2 ./kmeans_mpi vectors.bin 10 50 1
mpirun --oversubscribe -np 4 ./kmeans_mpi vectors.bin 10 50 1
# Output: cluster_labels_mpi.bin (overwritten each run)

# MPI+OMP hybrid
mpirun --oversubscribe -np 2 ./kmeans_mpi vectors.bin 10 50 2
mpirun --oversubscribe -np 2 ./kmeans_mpi vectors.bin 10 50 4
mpirun --oversubscribe -np 4 ./kmeans_mpi vectors.bin 10 50 2
mpirun --oversubscribe -np 4 ./kmeans_mpi vectors.bin 10 50 4
```

11.4 Verify Correctness

```
python3 -c "
import struct
def read(f):
    with open(f,'rb') as fp:
        n=struct.unpack('i',fp.read(4))[0]
        return list(struct.unpack(str(n)+'i',fp.read(n*4)))
s,o,m=read('cluster_labels.bin'),read('cluster_labels_omp.bin'),\
    read('cluster_labels_mpi.bin')
```

```
print('serial==omp:',s==o,'| serial==mpi:',s==m)
print('diff omp:',sum(a!=b for a,b in zip(s,o)),
      '| diff mpi:',sum(a!=b for a,b in zip(s,m)))
"
# Expected: serial==omp: True | serial==mpi: True
#           diff omp: 0 | diff mpi: 0
```

11.5 Generate Plots and Quality Report

```
# Regenerate all three plots from benchmark_results.csv
python3 plot_scaling.py

# Evaluate clustering quality (Purity / NMI / ARI vs ground truth)
python3 evaluate.py
```

12 Conclusions

This project demonstrates a complete end-to-end HPC scaling study of K-Means clustering on real high-dimensional data. Key findings from the actual measured results are:

1. **Correctness.** All three parallel variants produce bit-identical results (0 label mismatches out of 50,000). All converge at iteration 47 with inertia 1.2479×10^6 .
2. **Best raw speedup: OMP 8T.** Pure OpenMP at 8 threads achieves $4.19\times$ speedup (1.470 s vs. 6.159 s serial), outperforming all MPI and hybrid configurations on this single-node WSL machine.
3. **Best hybrid: MPI+OMP 4R $\times$ 2T.** The hybrid configuration achieves 2.161 s ($2.85\times$ speedup) — the best among all multi-process configurations, confirming that combining MPI decomposition with intra-rank OMP threads can outperform pure MPI, even on a single node.
4. **OMP outperforms MPI on single node.** OpenMP avoids the process-scheduling overhead of MPI in a WSL environment, giving better scaling per unit resource for intra-node workloads.
5. **Oversubscription causes severe regression.** MPI+OMP 4R $\times$ 4T (16 virtual processors) runs at 4.841 s in the manual re-run — slower than serial — due to OS thread oversubscription. This result underscores that adding processors beyond the physical core count actively hurts performance.
6. **Clustering quality is semantically meaningful.** Purity = 0.536, NMI = 0.441, ARI = 0.321 confirm that MobileNetV2 features capture genuine semantic structure. Truck (77.0%) is most distinct; Bird (26.8%) is hardest due to high intra-class visual variance.
7. **Amdahl's Law.** The parallelisable fraction is estimated at $\approx 72\%$ (hybrid) to $\approx 84\%$ (OMP), capping theoretical maximum speedup. Further gains require either parallelising the serial K-Means++ initialisation or running on a true multi-node cluster where distributed MPI is justified.